

19th day: Numpy operation, Masking or filter in matrix & OTP Genration,

```
In [9]: import numpy as np
m= np.random.randint(15,150,(10,10))
print(m)
```

```
[[ 98  51 131  65 104  39  78  66  55  43]
 [130 137 107  26  35  41  92  36  78  24]
 [119 140  71 118  96  93  62  98 112 137]
 [107  26  98  21  46 118  17 107 145  44]
 [149  63  15  74 117 109  79  15  51 120]
 [137  38  53 132 122 145  93  96  79 142]
 [ 26 117  43  68  92  77  37  67 107  52]
 [ 52  51  97 131  82 124  81  29  17 100]
 [ 87  16  18  80  85  36 126  53 113  77]
 [ 63 141 112  40  16  16  47  38  39 134]]
```

```
In [68]: np.arange(1,13).reshape(12,1)
```

```
Out[68]: array([[ 1],
 [ 2],
 [ 3],
 [ 4],
 [ 5],
 [ 6],
 [ 7],
 [ 8],
 [ 9],
 [10],
 [11],
 [12]])
```

```
In [69]: b = np.random.randint(10,20,(5,4))
b
```

```
Out[69]: array([[11, 13, 12, 17],
 [10, 10, 15, 10],
 [15, 10, 12, 17],
 [19, 11, 17, 14],
 [16, 14, 16, 11]], dtype=int32)
```

```
In [70]: type(b)
```

```
Out[70]: numpy.ndarray
```

```
In [71]: b
```

```
Out[71]: array([[11, 13, 12, 17],
               [10, 10, 15, 10],
               [15, 10, 12, 17],
               [19, 11, 17, 14],
               [16, 14, 16, 11]], dtype=int32)
```

```
In [72]: b[1:3]
```

```
Out[72]: array([[10, 10, 15, 10],
               [15, 10, 12, 17]], dtype=int32)
```

```
In [73]: b
```

```
Out[73]: array([[11, 13, 12, 17],
               [10, 10, 15, 10],
               [15, 10, 12, 17],
               [19, 11, 17, 14],
               [16, 14, 16, 11]], dtype=int32)
```

```
In [74]: b[0:-2]
```

```
Out[74]: array([[11, 13, 12, 17],
               [10, 10, 15, 10],
               [15, 10, 12, 17]], dtype=int32)
```

```
In [75]: b
```

```
Out[75]: array([[11, 13, 12, 17],
               [10, 10, 15, 10],
               [15, 10, 12, 17],
               [19, 11, 17, 14],
               [16, 14, 16, 11]], dtype=int32)
```

```
In [76]: b[-5,-3]
```

```
Out[76]: np.int32(13)
```

```
In [77]: b[-4,2]
```

```
Out[77]: np.int32(15)
```

```
In [10]: np.arange(15,150)
```

```
Out[10]: array([ 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,
                28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40,
                41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53,
                54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66,
                67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79,
                80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92,
                93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105,
                106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118,
                119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131,
                132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144,
                145, 146, 147, 148, 149])
```

```
In [11]: m= np.random.randint(15,150,(10,10))
print(m)
```

```
[[ 29  16  30 139  46 146 100  85  40 124]
 [116 136  50  27 128 115  74  22  32  39]
 [ 17 142  27 135 104  53 144 122  66  74]
 [ 79 140  89 113  47  47  76 116 135 110]
 [ 76 101  35  91 149  42  56  20  68  88]
 [144 101  96  67  34  33 105  60  46  50]
 [105  72  63 101 131  32  21  50  53  82]
 [ 75  68  59  79  28  36  80 109 115  21]
 [145  27  83  30  31  32  43 149  26  89]
 [108  71  27 125  37  27  81 134 125 113]]
```

```
In [12]: m[0]
```

```
Out[12]: array([ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124], dtype=int32)
```

```
In [13]: m[:]
```

```
Out[13]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
 [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
 [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
 [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
 [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
 [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
 [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
 [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
 [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
 [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [14]: m[::-1]
```

```
Out[14]: array([[108,  71,  27, 125,  37,  27,  81, 134, 125, 113],
 [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
 [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
 [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
 [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
 [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
 [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
 [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
 [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
 [ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124]], dtype=int32)
```

```
In [15]: m[:,2]
```

```
Out[15]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
 [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
 [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
 [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
 [145,  27,  83,  30,  31,  32,  43, 149,  26,  89]], dtype=int32)
```

```
In [16]: m[:,3]
```

```
Out[16]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
               [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
               [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [17]: m[0,3]
```

```
Out[17]: np.int32(139)
```

```
In [20]: m[0:2:3]
```

```
Out[20]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124]], dtype=int32)
```

```
In [23]: m[:,]
```

```
Out[23]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
               [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
               [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
               [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
               [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
               [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
               [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
               [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
               [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [24]: m[2,4]
```

```
Out[24]: np.int32(104)
```

```
In [26]: m[2:4]      # use slicing formula (n-1) for 2nd value
```

```
Out[26]: array([[ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110]], dtype=int32)
```

```
In [27]: m[:,0]
```

```
Out[27]: array([ 29, 116,  17,  79,  76, 144, 105,  75, 145, 108], dtype=int32)
```

```
In [28]: m[:, -3]
```

```
Out[28]: array([ 85,  22, 122, 116,  20,  60,  50, 109, 149, 134], dtype=int32)
```

```
In [33]: m
```

```
Out[33]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
               [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
               [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
               [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
               [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
               [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
               [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
               [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
               [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [34]: m[4:]
```

```
Out[34]: array([[ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
               [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
               [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
               [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
               [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
               [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [35]: m
```

```
Out[35]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
               [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
               [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
               [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
               [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
               [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
               [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
               [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
               [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [36]: m[:4]
```

```
Out[36]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
               [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
               [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110]], dtype=int32)
```

```
In [37]: m
```

```
Out[37]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
               [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
               [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
               [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
               [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
               [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
               [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
               [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
               [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [39]: m[1:4,1:3] # it print specific value of [1:4]for row part,[1:3] for column
```

```
Out[39]: array([[136,  50],
               [142,  27],
               [140,  89]], dtype=int32)
```

```
In [40]: m
```

```
Out[40]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
               [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
               [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
               [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
               [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
               [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
               [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
               [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
               [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [41]: m[3:8,3:8]
```

```
Out[41]: array([[113,  47,  47,  76, 116],
               [ 91, 149,  42,  56,  20],
               [ 67,  34,  33, 105,  60],
               [101, 131,  32,  21,  50],
               [ 79,  28,  36,  80, 109]], dtype=int32)
```

```
In [78]: m
```

```
Out[78]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
               [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
               [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
               [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
               [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
               [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
               [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
               [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
               [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [79]: m[:,8]
```

```
Out[79]: array([ 40,  32,  66, 135,  68,  46,  53, 115,  26, 125], dtype=int32)
```

```
In [82]: m
```

```
Out[82]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
               [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
               [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
               [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
               [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
               [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
               [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
               [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
               [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
               [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [84]: m[:6]
```

```
Out[84]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
                [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
                [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
                [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
                [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
                [144, 101,  96,  67,  34,  33, 105,  60,  46,  50]], dtype=int32)
```

```
In [86]: m[:,8]
```

```
Out[86]: array([ 40,  32,  66, 135,  68,  46,  53, 115,  26, 125], dtype=int32)
```

```
In [87]: m[:, -1]
```

```
Out[87]: array([124,  39,  74, 110,  88,  50,  82,  21,  89, 113], dtype=int32)
```

```
In [88]: m[3:-3]
```

```
Out[88]: array([[ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
                [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
                [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
                [105,  72,  63, 101, 131,  32,  21,  50,  53,  82]], dtype=int32)
```

```
In [89]: m[6]
```

```
Out[89]: array([105,  72,  63, 101, 131,  32,  21,  50,  53,  82], dtype=int32)
```

```
In [90]: m[:6]
```

```
Out[90]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
                [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
                [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
                [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
                [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
                [144, 101,  96,  67,  34,  33, 105,  60,  46,  50]], dtype=int32)
```

```
In [91]: m[5:7]
```

```
Out[91]: array([[144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
                [105,  72,  63, 101, 131,  32,  21,  50,  53,  82]], dtype=int32)
```

```
In [92]: m
```

```
Out[92]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
                [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
                [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
                [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
                [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
                [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
                [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
                [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
                [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
                [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```



```
In [93]: m[0:10:3]
```

```
Out[93]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
                [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
                [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
                [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [94]: m[4:30:2]
```

```
Out[94]: array([[ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
                [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
                [145,  27,  83,  30,  31,  32,  43, 149,  26,  89]], dtype=int32)
```

```
In [95]: m[:, :-2]
```

```
Out[95]: array([[108,  71,  27, 125,  37,  27,  81, 134, 125, 113],
                [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
                [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
                [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
                [116, 136,  50,  27, 128, 115,  74,  22,  32,  39]], dtype=int32)
```

```
In [96]: m[:, 50]
```

```
Out[96]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
                [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
                [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
                [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
                [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
                [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
                [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
                [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
                [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
                [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [97]: m[-10:]
```

```
Out[97]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
                [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
                [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
                [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
                [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
                [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
                [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
                [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
                [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
                [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [99]: m[:, -5]
```

```
Out[99]: array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
                [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
                [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
                [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
                [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88]], dtype=int32)
```


In [100...

m

Out[100...

```
array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
       [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
       [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
       [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
       [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
       [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
       [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
       [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
       [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
       [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

In [101...

m[2:6,2:4]

Out[101...

```
array([[ 27, 135],
       [ 89, 113],
       [ 35,  91],
       [ 96,  67]], dtype=int32)
```

In [105...

m[8:10,6:10]

Out[105...

```
array([[ 43, 149,  26,  89],
       [ 81, 134, 125, 113]], dtype=int32)
```

In [106...

m[1:2,2:4]

Out[106...

```
array([[50, 27]], dtype=int32)
```

In [107...

m

Out[107...

```
array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
       [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
       [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
       [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
       [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
       [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
       [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
       [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
       [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
       [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

In [108...

m[2:3,2:3]

Out[108...

```
array([[27]], dtype=int32)
```

In [109...

m

```
Out[109...] array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
        [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
        [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
        [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
        [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
        [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
        [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
        [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
        [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
        [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [110...] m[2:4,3:5]
```

```
Out[110...] array([[135, 104],
        [113,  47]], dtype=int32)
```

```
In [111...] m[3:5,2:4]
```

```
Out[111...] array([[ 89, 113],
        [ 35,  91]], dtype=int32)
```

```
In [112...] m
```

```
Out[112...] array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
        [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
        [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
        [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
        [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
        [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
        [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
        [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
        [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
        [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [113...] m[3:5,2:5]
```

```
Out[113...] array([[ 89, 113,  47],
        [ 35,  91, 149]], dtype=int32)
```

Masking or Filter in matrix

```
In [114...] m
```

```
Out[114...] array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
        [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
        [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
        [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
        [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
        [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
        [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
        [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
        [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
        [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [115... m.shape
```

```
Out[115... (10, 10)
```

```
In [46]: len(m)
```

```
Out[46]: 10
```

```
In [47]: type(m)
```

```
Out[47]: numpy.ndarray
```

```
In [116... id(m)
```

```
Out[116... 2044147921520
```

```
In [117... m
```

```
Out[117... array([[ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124],
        [116, 136,  50,  27, 128, 115,  74,  22,  32,  39],
        [ 17, 142,  27, 135, 104,  53, 144, 122,  66,  74],
        [ 79, 140,  89, 113,  47,  47,  76, 116, 135, 110],
        [ 76, 101,  35,  91, 149,  42,  56,  20,  68,  88],
        [144, 101,  96,  67,  34,  33, 105,  60,  46,  50],
        [105,  72,  63, 101, 131,  32,  21,  50,  53,  82],
        [ 75,  68,  59,  79,  28,  36,  80, 109, 115,  21],
        [145,  27,  83,  30,  31,  32,  43, 149,  26,  89],
        [108,  71,  27, 125,  37,  27,  81, 134, 125, 113]], dtype=int32)
```

```
In [48]: m<50      # all value in list
```

```
Out[48]: array([[ True,  True,  True, False,  True, False, False, False,  True,
        False],
        [False, False, False,  True, False, False, False,  True,  True,
        True],
        [ True, False,  True, False, False, False, False, False, False,
        False],
        [False, False, False, False,  True,  True, False, False, False,
        False],
        [False, False,  True, False, False,  True, False,  True, False,
        False],
        [False, False, False, False,  True,  True, False, False,  True,
        False],
        [False, False, False, False, False,  True,  True, False, False,
        False],
        [False, False, False, False,  True,  True, False, False, False,
        True],
        [False,  True, False,  True,  True,  True,  True, False,  True,
        False],
        [False, False,  True, False,  True,  True, False, False, False,
        False]])
```

```
In [49]: m>50
```

```
Out[49]: array([[False, False, False,  True, False,  True,  True,  True, False,
                True],
               [ True,  True, False, False,  True,  True,  True, False, False,
                False],
               [False,  True, False,  True,  True,  True,  True,  True,  True,
                True],
               [ True,  True,  True,  True, False, False,  True,  True,  True,
                True],
               [ True,  True, False,  True,  True, False,  True, False,  True,
                True],
               [ True,  True,  True,  True, False, False,  True,  True, False,
                False],
               [ True,  True,  True,  True,  True, False, False, False,  True,
                True],
               [ True,  True,  True,  True, False, False,  True,  True,  True,
                False],
               [ True, False,  True, False, False, False, False,  True, False,
                True],
               [ True,  True, False,  True, False, False,  True,  True,  True,
                True]])
```

```
In [50]: m[m<50]
```

```
Out[50]: array([29, 16, 30, 46, 40, 27, 22, 32, 39, 17, 27, 47, 47, 35, 42, 20, 34,
                33, 46, 32, 21, 28, 36, 21, 27, 30, 31, 32, 43, 26, 27, 37, 27],
                dtype=int32)
```

```
In [52]: m[m>50]
```

```
Out[52]: array([139, 146, 100,  85, 124, 116, 136, 128, 115,  74, 142, 135, 104,
                53, 144, 122,  66,  74,  79, 140,  89, 113,  76, 116, 135, 110,
                76, 101,  91, 149,  56,  68,  88, 144, 101,  96,  67, 105,  60,
                105,  72,  63, 101, 131,  53,  82,  75,  68,  59,  79,  80, 109,
                115, 145,  83, 149,  89, 108,  71, 125,  81, 134, 125, 113],
                dtype=int32)
```

```
In [54]: m[m!=50]
```

```
Out[54]: array([ 29,  16,  30, 139,  46, 146, 100,  85,  40, 124, 116, 136,  27,
                128, 115,  74,  22,  32,  39,  17, 142,  27, 135, 104,  53, 144,
                122,  66,  74,  79, 140,  89, 113,  47,  47,  76, 116, 135, 110,
                76, 101,  35,  91, 149,  42,  56,  20,  68,  88, 144, 101,  96,
                67,  34,  33, 105,  60,  46, 105,  72,  63, 101, 131,  32,  21,
                53,  82,  75,  68,  59,  79,  28,  36,  80, 109, 115,  21, 145,
                27,  83,  30,  31,  32,  43, 149,  26,  89, 108,  71,  27, 125,
                37,  27,  81, 134, 125, 113], dtype=int32)
```

```
In [64]: m[m==50]
```

```
Out[64]: array([50, 50, 50], dtype=int32)
```

```
In [118... m==50
```

```
Out[118...] array([[False, False, False, False, False, False, False, False, False,
False],
[False, False, True, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
True],
[False, False, False, False, False, False, False, True, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False]])
```

```
In [63]: m[m>=50]
```

```
Out[63]: array([139, 146, 100, 85, 124, 116, 136, 50, 128, 115, 74, 142, 135,
104, 53, 144, 122, 66, 74, 79, 140, 89, 113, 76, 116, 135,
110, 76, 101, 91, 149, 56, 68, 88, 144, 101, 96, 67, 105,
60, 50, 105, 72, 63, 101, 131, 50, 53, 82, 75, 68, 59,
79, 80, 109, 115, 145, 83, 149, 89, 108, 71, 125, 81, 134,
125, 113], dtype=int32)
```

```
In [62]: m[m<=50]
```

```
Out[62]: array([29, 16, 30, 46, 40, 50, 27, 22, 32, 39, 17, 27, 47, 47, 35, 42, 20,
34, 33, 46, 50, 32, 21, 50, 28, 36, 21, 27, 30, 31, 32, 43, 26, 27,
37, 27], dtype=int32)
```

```
In [120...] A1 = m[m<50]
A1
```

```
Out[120...] array([29, 16, 30, 46, 40, 27, 22, 32, 39, 17, 27, 47, 47, 35, 42, 20, 34,
33, 46, 32, 21, 28, 36, 21, 27, 30, 31, 32, 43, 26, 27, 37, 27],
dtype=int32)
```

```
In [121...] A2 = m[m>50]
A2
```

```
Out[121...] array([139, 146, 100, 85, 124, 116, 136, 128, 115, 74, 142, 135, 104,
53, 144, 122, 66, 74, 79, 140, 89, 113, 76, 116, 135, 110,
76, 101, 91, 149, 56, 68, 88, 144, 101, 96, 67, 105, 60,
105, 72, 63, 101, 131, 53, 82, 75, 68, 59, 79, 80, 109,
115, 145, 83, 149, 89, 108, 71, 125, 81, 134, 125, 113],
dtype=int32)
```

```
In [122...] A3 = m[m<=50]
A3
```

```
Out[122...] array([29, 16, 30, 46, 40, 50, 27, 22, 32, 39, 17, 27, 47, 47, 35, 42, 20,
      34, 33, 46, 50, 32, 21, 50, 28, 36, 21, 27, 30, 31, 32, 43, 26, 27,
      37, 27], dtype=int32)
```

```
In [123...] A4 = m[m==50]
A4
```

```
Out[123...] array([50, 50, 50], dtype=int32)
```

```
In [127...] A5 = m[m<100]
A5
```

```
Out[127...] array([29, 16, 30, 46, 85, 40, 50, 27, 74, 22, 32, 39, 17, 27, 53, 66, 74,
      79, 89, 47, 47, 76, 76, 35, 91, 42, 56, 20, 68, 88, 96, 67, 34, 33,
      60, 46, 50, 72, 63, 32, 21, 50, 53, 82, 75, 68, 59, 79, 28, 36, 80,
      21, 27, 83, 30, 31, 32, 43, 26, 89, 71, 27, 37, 27, 81],
      dtype=int32)
```

```
In [128...] A5 = m[m>100]
A5
```

```
Out[128...] array([139, 146, 124, 116, 136, 128, 115, 142, 135, 104, 144, 122, 140,
      113, 116, 135, 110, 101, 149, 144, 101, 105, 105, 101, 131, 109,
      115, 145, 149, 108, 125, 134, 125, 113], dtype=int32)
```

```
In [130...] A5 = m[m==100]
A5
```

```
Out[130...] array([100], dtype=int32)
```

```
In [131...] A6 = m[m<=100]
A6
```

```
Out[131...] array([ 29,  16,  30,  46, 100,  85,  40,  50,  27,  74,  22,  32,  39,
      17,  27,  53,  66,  74,  79,  89,  47,  47,  76,  76,  35,  91,
      42,  56,  20,  68,  88,  96,  67,  34,  33,  60,  46,  50,  72,
      63,  32,  21,  50,  53,  82,  75,  68,  59,  79,  28,  36,  80,
      21,  27,  83,  30,  31,  32,  43,  26,  89,  71,  27,  37,  27,
      81], dtype=int32)
```

```
In [132...] A7 = m[m>=100]
A7
```

```
Out[132...] array([139, 146, 100, 124, 116, 136, 128, 115, 142, 135, 104, 144, 122,
      140, 113, 116, 135, 110, 101, 149, 144, 101, 105, 105, 101, 131,
      109, 115, 145, 149, 108, 125, 134, 125, 113], dtype=int32)
```

```
In [133...] A8 = m[m!=100]
A8
```

```
Out[133...] array([ 29, 16, 30, 139, 46, 146, 85, 40, 124, 116, 136, 50, 27,
      128, 115, 74, 22, 32, 39, 17, 142, 27, 135, 104, 53, 144,
      122, 66, 74, 79, 140, 89, 113, 47, 47, 76, 116, 135, 110,
      76, 101, 35, 91, 149, 42, 56, 20, 68, 88, 144, 101, 96,
      67, 34, 33, 105, 60, 46, 50, 105, 72, 63, 101, 131, 32,
      21, 50, 53, 82, 75, 68, 59, 79, 28, 36, 80, 109, 115,
      21, 145, 27, 83, 30, 31, 32, 43, 149, 26, 89, 108, 71,
      27, 125, 37, 27, 81, 134, 125, 113], dtype=int32)
```

```
In [138...] A9 = m[m<150]
A9
```

```
Out[138...] array([ 29, 16, 30, 139, 46, 146, 100, 85, 40, 124, 116, 136, 50,
      27, 128, 115, 74, 22, 32, 39, 17, 142, 27, 135, 104, 53,
      144, 122, 66, 74, 79, 140, 89, 113, 47, 47, 76, 116, 135,
      110, 76, 101, 35, 91, 149, 42, 56, 20, 68, 88, 144, 101,
      96, 67, 34, 33, 105, 60, 46, 50, 105, 72, 63, 101, 131,
      32, 21, 50, 53, 82, 75, 68, 59, 79, 28, 36, 80, 109,
      115, 21, 145, 27, 83, 30, 31, 32, 43, 149, 26, 89, 108,
      71, 27, 125, 37, 27, 81, 134, 125, 113], dtype=int32)
```

```
In [140...] A10 = m[m<40]
A10
```

```
Out[140...] array([29, 16, 30, 27, 22, 32, 39, 17, 27, 35, 20, 34, 33, 32, 21, 28, 36,
      21, 27, 30, 31, 32, 26, 27, 37, 27], dtype=int32)
```

Python program to generate otp

```
In [143...] import random
```

```
In [146...] def generate_otp(length=6):
    """Generate a numeric OTP of a specified length."""
    digits = '012345'
    otp = ''.join(random.choice(digits) for _ in range(length))
    return otp
```

Example usage

```
In [149...] otp_length = 6      # You can change this to any length you prefer
otp = generate_otp(otp_length)
print(f"Your OTP is:{otp}")
```

Your OTP is:403115

created by own

```
In [150...] def generate_otp(length=4):
    """Generate a numeric and alphabet OTP of a specified length."""
    digits = '1a2b'
```



```
otp = ''.join(random.choice(digits) for _ in range(length))
return otp
```

```
In [158... otp_length = 4
otp = genrate_otp(otp_length)
print(f"Your OTP is : {otp}")
```

Your OTP is : b12b

```
In [160... def wish():
    print('good even')
    wish()

def wish():
    print('good even')
    wish()

def wish():
    print('good even')
    wish()

def wish():
    print('good even')
    wish()
    wish()
    wish()
```

```
In [161... list1=['a', 'b', 'c', 1,6]
print(list1.pop)
```

<built-in method pop of list object at 0x000001DBF4398DC0>

```
In [162... x = [1,2,3,4]
y = x.copy()
x.append(4)
print(x)
```

[1, 2, 3, 4, 4]

In []:

In []:

In []:

In []:

In []: